# L12.3: The Hessian matrix and local extrema for f(x,y,z)

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online BSc program on data science

and programming. This video is about the Hessian matrix and local extrema for f x y z. In the

previous video we have seen the Hessian test to classify critical points for the function

f x y. In this video, we will see the same or an analogous test for f x y z.

So, as you can see that as your number of variables increases, the test sort of becomes

a bit harder, which is why we have these separate videos. Now, one can make sense of all this

in terms of linear algebra for the Hessian matrix for n variables. And in particular,

if we allow ourselves the notions of Eigen values, which you will study later in your

machine learning course, and also use these ideas of diagonalization and orthogonal, using

orthogonal matrices. So, we did this when we did similarity and equivalence of matrices.

Using all those ideas, we can formulate a general test for n variables. But in the interest

of time, and functionality, as far as this course is concerned, I am not doing this for

general variables, although we have developed some of the linear algebra already required

for this, but we need a little bit more. So, I am restricting myself to 3 variables. And

this will be you enough probably for most of the problems that you will come across.

So, let us recall what is the Hessian test for a function of 2 variables. So, if you

have a function defined on a domain D in R2, and a is a critical point such that the first

and second order partial derivatives are continuous in an open ball around a, then there is something

called the Hessian test, which means that you compute the Hessian matrix of the function

f and evaluated at the point a, that gives you a 2 by 2 matrix.

You compute its determinant and the value of the 11 term. If that determinant is positive,

and the 1 1 term is also positive, then the point is a local minimum. If the determinant

is positive, and the one-oneth term, meaning the fxx at a tilde is negative, then it is

a local maximum. If the determinant is negative, it is a saddle point. And if the determinant

is 0, which is called the degenerate case.

So, the previous 3 are called the non-degenerate cases, which means that the determinant is

non-zero. But in the degenerate case, which is to say the Hessian determinant is 0, this

test is inconclusive. So, that is really the bad case. That is the case we want to avoid,

because that means that the test does not tell us anything. So, we saw some examples

of this test, we also saw that it told us the answer in some surprising cases.

Whereas, in some cases where it was obvious what the answer should be, it turned out to

be inconclusive. So, the test has a, it comes with, it is a double-edged test, sometimes

it will be it will work in situations where you really do not know what is happening.

And sometimes it will not work, even though you absolutely know what is happening. So,

we will see something similar for the Hessian test in three variables.

I am not going to do as many examples here. Because we already know the phenomenon that

are taking place, namely that it can work sometimes in spite of the fact that we have

no clue of what is happening. And it may not work at all, despite us knowing what is happening.

So, let us see what the test is. So, again, this is a function f is a function defined

on a domain D in 3 variables. Suppose a is a critical point and the first and second

partial derivatives are a continuous in an open ball around a tilde. I will remind you

that now there are 9 second order partial derivatives fxx, fxy, fxz, fyx, fyy, fyz and

then fzx, fzy, fzz. Of course, thanks to Clairaut's theorem, we know that we can interchange the

order.

So, fxy and fx yx are the same, fxz and fzx are the same, and fyz and fzy are the same.

So, effectively, there are 6 partial derivatives. So, this is a fairly long computation if we

want to compute this, and then the Hessian matrix is where we put these second order

partial derivatives into a 3 by 3 matrix, remember, it is a 3 by 3 matrix now, because

you have 3 variables and then the test tells us that if the following things happen, then

we can classify the critical points.

So, if the term fxx is positive, the term fxx times fyy minus f squared xy, which was

the determinant that we used in the 2 by 2 case if that at a tilde is positive, and if

the determinant of the entire Hessian matrix is positive, then a is a local minimum. If

on the other hand, the fxx is negative, the second term fxx fyy minus f squared xy at

a tilde is positive and the determinant is negative. So, this is alternating sign, then

a tilde is a local maximum.

If the determinant is non-zero, and neither of cases 1 or 2 occur, then it is a saddle

point. So, that means in these two, the first two cases we have local minimum or local maximum,

and in all the other non-degenerate cases, meaning where the determinant is non-zero,

it is a saddle point. And if the determinant is 0, which means it is a degenerate situation,

then the test is inconclusive. So, this is the no Hessian test in the three variables

situation.

This is getting slightly complicated as you can see, and which is why we, I am not describing

this in the general n by n setup. It actually has an easier description, but for that, we

need to know what are Eigen values.

So, let us understand these terms better. So, the terms involved in the test are fxx,

fxx fyy minus f squared x, y at a tilde and determinant of this matrix at a tilde. So,

what are these terms? Let us write down the matrix. So, here is the matrix fxx fxy, fxz,

so I am not really caring now about the order whether it is fxy or fyx because of Clairaut's

theorem, fyy, fyz and then fxz, fyz and then fzz and let me draw this like this. So, this

is our hessian matrix. So, the Hessian matrix and maybe let us do this at a tilde so in

that case, I would evaluate all these at a tilde.

So, once I evaluate, these are numbers, these are all numbers. And what do these terms correspond.

So, fxx at a tilde corresponds to this term there, so it is the one-oneth term or the

1 by 1 matrix in the top left corner. This second term. So, the second term here, this

term so, this term corresponds to the determinant of the 2 by 2 matrix in the top left corner.

And the third term which is the determinant of Hf of a tilde corresponds to the determinant

of the entire matrix.

So, what we are doing is, we are taking the determinants of the top left corner successively

larger. So, the 1 by 1, 2 by 2, 3 by 3 and in fact, we can generalize this to n by n,

only thing is those signs, we have to be careful of how they work. And what we are saying is

that if that top left corner, if the sign is like this, if it is plus plus and plus,

so which means that this is the 1 by 1, 2 by 2, 3 by 3. If it is plus plus and plus,

that means they are greater than 0, greater than 0 and greater than 0, then this is a

local minimum.

If it is a minus plus and a minus note, the alternating thing, this is very, very important.

This is where students often make mistakes. If it is a minus plus and a minus, it is a

local maximum and in all the other cases other cases, where this is not 0, where determinant

is not 0, all other non-degenerate cases by which we mean that the determinant is not

zero, this is a saddle point. And if it is the de degenerate case, which means determinant

is 0. This, we cannot really say this is the really bad, bad part. So, this is what it

seems.

So, what you have to keep track of is you have to compute 3 successive determinants,

the first one is a 1 by 1 determinant. So, that really there is nothing to say. The second

one is a 2 by 2 determinant. And the third one is a 3 by 3 determinant. So, in practice,

you do it the other way. Because you want to check that the determinant of the entire

matrix first is non-zero, if the determinant of the entire matrix is 0, then you stop the

test right there.

So, what you do is you your testing should start in the opposite direction. So, you do

your testing in this direction. First, you test this, then you test this, and then you

test this right. So, first, you test the entire matrix, what is the determinant, whether it

is positive or negative, or 0. If it is 0, end of story. If it is positive or negative,

then you test for the 2 by 2, if the 2 by 2 by chance happens to be negative, stop right

there. Because as you can see, here, you want it to be positive, so then it is a saddle

point. If it is positive, then you check for the 1 by 1 that is the way to go.

So, let us do a couple of examples. So, these are again, sort of our test examples to remember

how the rules work. So, the functions are also very familiar here. So, let us first

do x square plus y square plus z square. Clearly this is a non-negative function, and the only

place it 0 is the point 0,0,0. So, this is, so 0,0,0 is a local minimum. And in particular,

in fact, it is a global minimum, and in particular, it is a local minimum.

So hopefully, the Hessian test will tell us that so gradient is 2x, 2y 2z. Hessian is,

well, you can compute this, now I am doing this rather fast 200, 020 and 002. Let us

compute the various things that we have to compute. So, at 0,0, so the critical point

is 0,0,0. And so, the Hessian at 0,0,0 is the same thing. And so, we should compute

the determinant of all the top left matrices of decreasing sides.

So, the determinant of the entire thing is 8, which is positive, actually, you can see

all of them are going to be positive, but I just determinant of the second one is that

0,sorry 20, 02. That is 4, positive. So, which, which matrix are we looking at? It is this

matrix here. And then determinant of the first, meaning the entry in the 1-1, one-oneth place,

that is this entry here, so that entry is positive. So, fxx at 0,0,0. So therefore,

this is a local 0,0,0 is a local minimum. And that is exactly what we had in our case

1. So, this is how you can remember case 1.

Similarly, if you take minus x squared minus y squared minus z squared by the same argument,

it is always a non-positive function, the only place it 0 is the point 0,0,0. It is

a global maximum. And so hopefully, the Hessian test picks that out. So, I am going to sort

of do this really fast, gradient is minus 2x minus 2y minus 2z, the Hessian matrix now

you can write down is minus 2 0 0, 0 minus 2 0, 0 0 minus 2, critical point is clearly

0000. So, this is the Hessian also at that critical point.

And now let us see what the signs are. So, the determinant is of the entire matrix is

minus 2 times minus 2 times minus 2, it is a diagonal matrix, this is negative. So, we

have a minus sign, the determinant of the 2 by 2, that minus 2 0, 0 minus 2, that is

minus 2 times minus 2, it is 4, it is positive. And then the one-oneth entry is minus 2, which

is negative. So, the signs here are, this is the important part, what are the signs,

the signs are minus plus, minus, and that tells us this is a

local maximum. And indeed, this is what we already know and so this is how you can keep

this in mind.

Similarly, if you take x squared minus y squared plus z squared, you will get it is a saddle

point. So, I will just write down the Hessian and you can do the checking. So, the Hessian

is going to be 2 0 0, 0 minus 2 0; 0 0 2. So, the signs you are going to get are determinant

of the entire so the critical point is 0,0,0 and the signs you are going to get are so

minus 8, 2 by 2 is 2 0; 0 minus 2, which is minus 4 and already you can stop right here,

because Hessian is non, the Hessian determinant is non-zero and the signs are not corresponding

to either alternating or plus plus plus.

So, you can stop right here, but just to finish it at 000 this is for 2 which is positive.

So, this is a saddle point. Indeed, we already know this, because of, well, you can check

that along certain axis it is a local maximum, along certain axes it is a local minimum.

And finally, if you take x to the power 4 plus y to the power 4 plus z to the power

4 what happens the gradient is 4x cube 4y cube 4z cube, critical point is clearly 0,0,0.

By equating this to 0, and the Hessian you can compute from here is 12x squared 0 0,

0 12 y squared, 0, and 0 0 12 z squared. And what happens at the point, so at the point

0,0,0, this is the 0 matrix. And as a result, it is the degenerate case, we cannot say anything.

So, this is inconclusive, as far as the Hessian test is concerned.

On the other hand, we already know that x to the power 4 plus y to the power 4 plus

z to the power 4 is always non negative, it is a fourth power, they will always be non-negative.

And the only point where it takes the value 0 is the 0,0,0. So, this is in fact a global

minimum. So, in particular, a local minimum. So, in spite of the fact that we easily know

that this is a global minimum, the test is inconclusive. So, these are the 4 prototype

examples you should keep in mind to, in case you forget the tests. So, this will tell you.

Let us do a couple of examples of, of computing this. So, here is f x, y z is xy plus yz plus

zx. So, gradient f, we have actually computed the gradient and the Hessian, so the gradient

here is y plus z, z plus x, x plus y. So, you equate this to 0. So, then we get these

3 equations y plus z is 0, z plus x is 0, x plus y is 0. So, that tells us that x is

minus y, and are also tells us that x is z, that is what we get from equation 1 and 3,

but equation 2 tells us that x is minus z.

So, the only way this can happen is if x is equal y is equal to z is equal to 0 is equal

to 0. So, the only solutions are x is equal y is equal to z is equal to 0. So, the critical

point is 000. And what happens to the Hessian matrix? So, what is the Hessian, so if we

write down the Hessian, which we have actually done before, we get 0 1 1, 1 0 1 and 1 1 0.

So, this is in particular, the Hessian at 000. And what happens? So, let us do our test.

So first, what is the determinant of the hessian matrix? So, this is, so if we do our determinants,

this is 0 times determinant of the 1 by 1, 1 minor, minus 1 times determinant of 1 1,

1 0 plus 1 times determinant of 1 0, 1 1. So, if we compute this, this is 1 plus 1,

which is 2, so this is positive. So, at least it is non-zero. So, we are in business, this

test is going to be conclusive.

So, now we take the 2 by 2 left, top left matrix calculated determinant 0 1, 1 0. This

is minus 1 negative. And, well as soon as we know it is negative, this does not correspond

to case 1 or case 2. So, this means it is a saddle point. So therefore, 0,0,0 is a saddle

point. And I will encourage you to check why it is a saddle point. So, find a sequence

of points which go to 0,0,0 for which 0,0,0 on that curve, it is it is a minimum value,

and find some other curve on which it is a maximum 0,0,0 is a point of maximum. So, I

hope it is clear how we are using this test.

So, let us do this final example, for x to the power 4 plus y to the power 4 plus z to

the power 4 plus xyz. I want to do this example just to show you that sometimes these calculations

do become rather difficult, doable, but difficult. So, what is the gradient, so the gradient

is 4x cube, plus yz 4y cube plus xz 4z cube plus xy. Let us set this to 0. So, equating

to 0 and we have to really work this out, we get, so we have 3 equations. 4x cube plus

yz is 4y cube plus xz is 4z cube plus xy is 0.

So, now we have to solve these 3 equations. And note that these are not linear equations.

So, our linear algebra would not really help us over here. So, we have to solve sort of

see this and solve this by brute force. And also using the fact that, they have a very

nice form. So here, here is where your comfort with, with equations or algebra in general

will kick in.

So, case 1 is, there is one solution staring us in the face, namely, where x is equal to

y is equal to z is equal to 0. And, it is very important to note that, once one of them

is 0, all the others are 0. For example, if x is 0, then you can put that into the equation

4y cube plus xz is 0 and that will give you that 4y cube is 0 and that will say that y

is 0. And you can put it in the into the equation 4z cube plus xy is 0 and that will say that

z is 0. So, by symmetry, if one of them is 0, all of them are 0. So that is case 1.

The other cases where none of them are 0. So, all are non-zero, all are not non 0. So,

none of them are 0. So, in that case, what we can do is we can multiply each of our equations

suitably. So, the first equation we can multiply by x, and we can say that 4x to the power

4 plus xyz is 0, which means 4x to the power 4 is minus xyz but I can do the same thing

for the second equation by multiplying it by y and I can say that 4y to the power 4

plus xyz is 0, which is saying that 4y to the power 4 is minus xyz and by symmetry we

will also get that this is 4z to the power 4.

So, what that means is 4x to the power 4 is 4y to the power 4 is 4z to the power 4, so

x to the power 4 is y to the power 4 is z to the power 4 and they are not all 0. Fine

and now if we if we take the fourth root, what that will tell us is that x is plus or

minus y. And again, which is plus or minus z. And now we can, so, now we can substitute

this in into our equations. So, substitute in 4x cube plus yz is 0.

So, if you substitute this, what you are going to get is 4x cube plus x squared is 0. And

since x is non-zero, we can cancel x squared on both sides and what we get is x is either

you get this or 4x cube minus x squared is 0. And then what we get is x is 1 by 4 or

x is minus 1 by 4. And then so again, for y and z also, you get these three. Of course,

all possible solutions are may not be correct, because they have to satisfy these equations

over here.

So, then you have to back check for what values these are satisfied. And if you do that, you

will get your critical points as 0,0,0, which was case 1, and then from here, we will get

1 minus 1 by 4 comma 1 by 4 comma 1 by 4, 1 by 4 comma minus 1 by 4 comma 1 by 4, and

1 by 4 comma, 1 by 4 comma minus 1 by 4, and finally minus 1 by 4 comma minus 1 by 4 comma,

minus 1 by 4. So, these are the 5 critical points that you will get. And now we have

to check the nature of these critical points.

So, let us see what is the Hessian. So, if you compute the Hessian, what you are going

to get is 12x squared z, y, then z 12y squared, x, and then y 12 sorry, y x 12z squared. Fine,

so now we have to evaluate this Hessian at all these various points, and check what happens.

So, already, you can see that 0,0,0 this is inconclusive. So, we cannot really say anything

about 0,0,0. So, 0,0,0 it is inconclusive.

And again, I will encourage you to check what happens at 000. My claim is that this should

probably be a saddle point. What happens at the other points, and by symmetry, for these

three, at least, we should get the same thing and then we have to check minus 1 by 4 minus

1 by 4, minus 1 by 4. So, let us check minus 1 by 4 comma 1 by 4 comma 1 by 4. So, in this

case, the Hessian is, so the Hessian is, let us write that down, 3 by 4, 1 by 4, 1 by 4,

1 by 4, 3 by 4, minus 1 by 4, and then 1 by 4, minus 1 by 4, 3 by 4.

Now, you can use, use our good old determinants, and check what we are getting. So, notice

here that there are constants in this, so you can pull out the constant. So, this matrix

is 1 by 4 multiplied by the matrix, which is consists of integers, if that makes your

computation easier. And remember that when you take determinant, since it is a 3 by 3

matrix, that constant will be to the power 3.

So, if you do that, the determinant of the entire thing is, I maybe, I not do the actual

computation and I will just say that this is positive you can see that is the case because

on the diagonal the terms are rather large. So, this is positive. And then for the 2 by

2, again, this is positive. And the 1 by 1 is again positive, positive, so this is a

plus plus plus situation that means this is a local. So, this is a all these 3 are local

minimum. And finally, we have the last point. And, again, actually the same thing happens.

So, in this case, you get 3 by 4, but all the off-diagonal entries are minus 1 by 4,

3 by 4 in the regular and minus 1 by 4 everywhere else. And if you compute the determinant,

again you are going to get plus plus plus, which means this is also a local minima. So,

once again, what what you will find here is that for the points, where it kind of not

so clear how the function behaves, the Hessian test is telling us something, whereas for

0,0,0, it is comparatively easier to see what is happening, because the function value there

is 0.

And you can try to sort of understand from which side do I come and along with curve

do I come, in order to get local maxima, local minima, etcetera. So, that behaves like a

certain point. So, the conclusion here is that is that again, the Hessian test is kind

of the behavior is kind of surprising, sometimes for things that you may understand relatively

better, it gives you inconclusive results, but for things that you may not understand

relating, you understand relatively less clearly, it gives you results.

Now, of course, we have not gone into proofs of the Hessian test. And maybe that is, that

is a separate topic in itself. So, we will not go there. There is a, we can extend this,

as I said, for the n by n case, but we will not do that in this course. So, the main theme

of this video was that we now have a Hessian test analogue to the second derivative test

for 3 variables. Thank you.